

Table 17: Example prompt for inverse modeling, chain-of-thought.

Inverse Modeling, CoT:

The following are two choices that people have made between different bags of candy. Each candy is a different color.

Choice 1 was made between the following bags:

Bag 1: red, brown, yellow, blue.

Bag 2: black.

The person making the choice chose Bag 2.

Choice 2 was made between the following bags:

Bag 1: yellow, black, red, brown.

The person making the choice chose Bag 1.

People were required to choose among the bags available, and were not allowed to reject all the bags.

For example, when there is only one bag, the person has no choice but to choose it.

Which choice (1 or 2) more strongly suggests that the person making the choice likes black candies?

Let's think step by step.

Table 18: List of 47 observed decisions from the inverse decision-making experiment of Jern et al. (2017). Decisions contained between 1-5 options, and each option corresponds to a column. The option in the leftmost column was chosen in all decisions. No options were empty; blank entries indicate that the decision had less than the maximum number of options. Each item is represented using a letter, with x being the target item that inferences are made upon.

option 1	option 2	option 3	option 4	option 5
d, c, b, a, x				
c, b, a, x				
b, a, x				
a, x				
x				
c, b, a, x	d, b, a, x			
a, x	b, x	c, x	d, x	
b, a, x	c, a, x			
b, a, x	b, c, x	b, d, x		
b, a, x	d, c, x			
a, x	b, x			
b, a, x	c, a, x	b, d, x		
a, x	b, x	c, x		
c, b, a, x	d			
b, a, x	c			
a, x	b			
b, a, x	c	d		
b, a, x	d, c			
a, x	b	c		
a, x	b, x	d, c		
b, a, x	b, d, c			
a, x	b, x	c, x	a, d	
a, x	b	c	d	
b, a, x	b, c, x	b, a, d		
a, x	b, x	a, c		
a, x	c, b			
c, b, a, x	c, b, a, d			
a, x	b	d, c		
a, x	b, x	a, c	a, d	
a, x	a, b			
b, a, x	b, a, c			
a, x	a, b	d, c		
a, x	d, c, b			
x	a			
b, a, x	b, a, c	b, a, d		
a, x	a, b	a, c		
a, x	a, b	a, c	a, d	
x	a	b		
x	a	b	c	
x	a	c, b		
x	a	b	c	d
x	c, b, a			
x	b, a	d, c		
x	a	b	d, c	
x	a	d, c, b		
x	d, c, b, a			